Auditing Closed-Loop Learning in Recurrent Neural Networks: Reproduction, Robustness, and Generalization

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Abstract

Recurrent neural networks often serve as mechanistic models of learning and control, but closed-loop training is difficult to reproduce because each action changes the inputs the model sees next. We independently implement the central protocol from Ger and Barak, first testing it with new seeds and protocol changes before extending the study to other architectures and tasks. In a double-integrator setup aligned with the original paper's main text, every one of 50 paired seeds exhibits a transient peak in the open-loop learner's deployed loss, with the mean peak reaching 19.1 times the initial loss. By the final epoch, losses are similar and no run reaches our 5 percent meaningful-gap threshold, so the trajectory through training is more informative than the endpoint. The reported spectral stages and coupled-stability transition also reproduce in all 50 seeds. In a targeted comparison of short and long horizons, all 120 runs meet the criterion combining short-horizon improvement with an increase in the spectral radius of the coupled agent-environment system; 62 also show long-horizon loss worsening, never without the radius increase. GRU controllers and tracking tasks retain final-loss divergence, and low-rank controllers meet the same criterion because their open-loop rollouts often produce extremely large losses. Tanh controllers retain only the transient peak, while final-loss transfer to path-integration tasks is weak. Increasing tanh hidden size from 50 to 400 preserves the peak and stability crossing, but a two-layer controller develops a persistent final gap. All 60 gated-controller and ring-task runs register a local-Jacobian crossing; the partial-observation variants finish close to the stability boundary and show weak behavioral transfer. These results motivate following deployed behavior throughout training with paired seeds, and also show that stability claims need coupled-system diagnostics and explicit reporting of horizons and failed runs.

1 Introduction

Closed-loop RNNs bring action-dependent feedback into sequence models and also serve as mechanistic models in computational neuroscience. During interaction, their outputs become part of later observations, creating a feedback path that is absent from open-loop imitation or teacher-forced learning. Ger and Barak (Ger and Barak, 2025) argued that this feedback changes how otherwise identical RNNs learn and attributed the important transitions to the joint dynamics of the controller and its environment.

We use their framework to retest individual claims about recurrent control with an independent implementation and new random seeds. We then vary the protocol and ask whether the coupled-system analysis extends to nonlinear controllers and path-integration tasks.

Explicit configuration files control our independent reproduction of the central open/closed-loop comparison, where criteria fixed before the reported sweeps turn qualitative claims about learning stages and stability into quantitative tests while retaining seed-level uncertainty. Later experiments identify protocol-sensitive results and motivate the reporting recommendations in Section 10.

2 Related Work

Task-trained RNNs in computational neuroscience. By solving cognitive or motor tasks, task-trained RNNs expose internal dynamics that can reveal low-dimensional mechanisms (Sussillo and Barak, 2013) and model context-dependent computation in prefrontal cortex (Mante et al., 2013). Barak describes these networks as hypothesis-generating tools in systems neuroscience (Barak, 2017); the fact that networks with similar performance can nevertheless have different representational geometry (Maheswaranathan et al., 2019) underscores why architecture and training choices matter, motivating our tests across protocols and model classes.

Dynamical-systems analysis of RNNs. The original paper belongs to a tradition of reverse-engineering trained RNNs through fixed points and local linearizations (Sussillo and Barak, 2013; Maheswaranathan et al., 2019). We study the RNN together with its environment because the environment turns actions into future observations, so the coupled dynamics may differ from those of the RNN alone.

Open-loop training, closed-loop deployment, and distribution shift. Teacher-forced training exposes a model to different states from those it reaches during deployment, a mismatch addressed by scheduled sampling in sequence prediction (Bengio et al., 2015) and by DAgger when the learner's own actions shift the visited states (Ross et al., 2011). Because open-loop RNN imitation can likewise minimize teacher-forced error and still fail after deployment, we evaluate every learner under closed-loop rollout.

Reproducibility and robustness in machine learning. The NeurIPS reproducibility program emphasized the value of artifacts and reporting checklists (Pineau et al., 2021), while work in deep reinforcement learning has shown how strongly random seeds and implementation or evaluation choices can affect empirical conclusions (Henderson et al., 2018). Closed-loop RNN experiments inherit these problems, with additional sensitivity to the rollout and to how the model is coupled to its environment.

We use *reproduction* for tests that preserve the original task family and claim. A *robustness* test changes the training or evaluation protocol, whereas a *generalization* test changes the architecture or task family; failure in either broader test narrows the claim's scope but does not by itself invalidate the original result.

3 Original Claims and Audit Questions

We distinguish the original paper's claims from the broader questions introduced by our audit. The three reproduction targets are the divergence between open- and closed-loop learning trajectories (C1), spectrally defined learning stages (C2), and stability of the coupled agent-environment system (C3).

Our two audit questions broaden the inquiry without attributing new claims to Ger and Barak: A1 examines robustness and the proposed tradeoff between short- and long-horizon performance, whereas A2 examines transfer across architectures and tasks.

We classify an effect as *reproduced* when its prespecified diagnostic holds for most paired seeds, requiring uncertainty intervals that exclude zero for continuous effects and using the per-seed threshold below for final-loss divergence. An effect is *partially reproduced* when seed-level support is unstable, a plausible perturbation changes the result, or diagnostic evidence is weak; *not reproduced* is reserved for effects that are absent or reversed, or that depend on hand-selected visual criteria.

For the C1 final-loss diagnostic, the smallest meaningful effect is a 5 percent relative deployed-loss gap, roughly 0.022 on the original-setting closed-loop scale; smaller gaps remain in the continuous results but do not count as final-loss divergence. Varying this threshold from 1 to 20 percent leaves every classification unchanged except for the partial-observation tanh ring, where 19/20 seeds exceed 1 percent and 2/20 exceed 5 percent (Appendix E). Because the original figure prominently shows a transient open-loop peak, we treat that peak as a separate C1 signature. Appendix C connects both C1 tests, along with the other decisions, to the original evidence.

4 Closed-Loop RNN Reproducibility Audit Protocol

We use this protocol as a reusable audit recipe for recurrent-control settings in which actions alter later observations. Each open-loop learner is paired with a closed-loop learner cloned from the same initialization, and criteria fixed before the reported sweeps evaluate the claimed stages and stability transition while retaining seed-level uncertainty. After the direct reproduction, we vary optimization and environment coupling before moving to held-out architectures and tasks. Coupled agent-environment spectra are computed whenever the joint dynamics are linear or locally linearizable; Appendix C gives the full protocol.

Each run, including failures, is classified on its own before the evidence is aggregated across seeds. This grounds claims about stages and mechanisms in seed-level measurements, so a few persuasive-looking curves cannot carry the conclusion.

4.1 Metrics and run accounting

Deployed closed-loop loss is the primary behavioral measure, with the remaining measures defined in Appendix B. The audit contains 675 completed paired runs: 50 in the core reproduction and 195 in the robustness sweep. The targeted A1 and A2 studies contribute 120 and 90, respectively, while another 220 runs support additional analyses. We also performed two one-seed protocol checks and five seeded executions of the original nonlinear notebook; all tables and figures are generated from saved outputs, and every non-finite outcome is recorded as a failure (F).

Detector thresholds were fixed during smoke-test development and remained unchanged throughout the reported sweeps. Across the sensitivity ranges in Appendix E, the claim decisions stay fixed at or above their prespecified values; in every A1 grid cell, all 120 runs meet both the combined short-horizon/radius criterion and the radius criterion itself, with no long-horizon loss worsening in the absence of a radius increase.

4.2 Paired open-loop and closed-loop training

For each seed, one initialization is cloned into an open-loop learner and a closed-loop learner. As in the public artifact, the closed-loop controller is trained by differentiating through the unrolled agent-environment dynamics, without a stochastic REINFORCE estimator. The open-loop model imitates teacher trajectories, and both models are evaluated under deployed closed-loop rollout.

4.3 Stage detection

The original paper defines its stages spectrally. During Stage 1, the effective position gain is negative and a complex-conjugate eigenvalue pair lies outside the unit circle; Stage 2 ends when the dominant eigenvalues of the coupled matrix enter the unit disk. Policy refinement follows in Stage 3 as the largest-magnitude real eigenvalue—the third mode in the original setting—grows. Our C2 detector follows this sequence, while loss derivatives and segmented changepoints are retained only as downstream observations.

4.4 Coupled-system stability diagnostics

For linearized settings, we compute the spectral radius, or largest eigenvalue magnitude, of both the recurrent matrix and the coupled agent-environment transition matrix. C3 predicts that the coupled radius will identify the stability transition more clearly than the recurrent-matrix radius.

4.5 Robustness and generalization

The A1 sweep changes optimization and environment coupling according to Table 4, with its Adam condition testing Ger and Barak's prediction that adaptive optimization mitigates the structured conflict between short and long horizons. A2 moves beyond the original setting by changing the controller class and then the task, including tracking and ring environments.

4.6 Original artifact, independent implementation, and deviations

We preserved the public artifact under `external/original_artifact/`, including its notebooks and data files. The reproduction log explains how to run each component and notes any data dependencies or execution constraints.

The reported experiments use a separate, configuration-controlled implementation of the full workflow. C1–C3 retain the original architecture and training profile, whereas A1 and A2 introduce new audit conditions. Every protocol choice comes from public materials and is documented in the configuration files; we did not contact the original authors. Notebook executions check the artifact itself, while all quantitative evidence comes from the independent implementation, whose commands for regenerating the tables and figures appear in Appendix A.

A code-level comparison confirms that the core reproduction follows the original double-integrator dynamics and training profile; Appendix D provides the item-by-item provenance. Where the public notebook differs from the main text, our aligned configuration sets $\beta = 0$ and samples $x_0 \sim [-2, 2]^2$, with squared Euclidean state cost averaged over the rollout. We treat this configuration as a claim-level reproduction of the protocol described in the main text.

Appendix D separates 22 exact artifact matches from five main-text-aligned deviations and ten new audit choices. Under the notebook protocol, the observed ranges from seeded executions of the original nonlinear notebook overlap those of the package runs for the final losses and peak statistics (Appendix H).

The open-loop input convention changes the interpretation of C1. White-noise teacher inputs cover the observation space more broadly and may therefore help imitation recover the teacher's full input-to-action map; under this convention, the transient peak is consistent with a partially learned policy passing through destabilizing parameter regions before its deployed loss approaches that of the closed-loop learner. Teacher-rollout inputs instead concentrate supervision on the teacher's own state distribution and leave deployment states uncorrected, matching the usual covariate-shift account in imitation learning (Ross et al., 2011; Bengio et al., 2015). In a controlled ablation, this convention produces a persistent final gap in all 15 seeds and no post-initial peak, with a median relative-gap ratio, $(L^{\mathrm{OL}} - L^{\mathrm{CL}})/L^{\mathrm{CL}}$, of approximately 147. All 65 comparable white-noise runs instead show the peak before finishing below the 5 percent final-gap threshold (Appendix G). Accordingly, C1 is evaluated using both the trajectory-level peak and the final deployed-loss gap.

5 Result 1: Direct Reproduction of the Core Divergence

Across 50 paired double-integrator runs, every open-loop student develops the post-initial deployed-loss peak shown in Figure 1. The peak occurs at a mean epoch of 249.1 (bootstrap 95 percent CI $[247.5, 250.5]$), close to the median of 249.5. On average, peak loss is 19.1 times the initial loss (CI $[17.1, 21.4]$) and 4.88×10^4 times the final loss. A learner can therefore look successful under teacher forcing even while its deployed closed-loop behavior deteriorates sharply during training.

After the peak, the open-loop learner recovers and the final losses become similar. The closed-loop learner finishes at 0.4370 (bootstrap 95 percent CI $[0.4367, 0.4373]$), while the open-loop learner finishes at 0.4387 (CI $[0.4383, 0.4390]$). Seed-to-seed variation is small: the respective SDs are 0.00109 and 0.00126, with IQRs $[0.4361, 0.4378]$ and $[0.4379, 0.4393]$. The paired final gap is positive but small, at 0.00168 (CI $[0.00147, 0.00190]$), and none of the 50 runs reaches the prespecified 5 percent threshold. C1 therefore reproduces through the learning trajectory and transient peak, with no meaningful separation at the endpoint.

All 50 core seeds also reproduce the spectral C2 sequence and the C3 stability transition: Stage 1 ends at a median epoch of 7, and Stage 2 ends with the stability crossing at a median epoch of 105.5. The final coupled spectral radius is 0.6253 (CI $[0.6243, 0.6264]$).

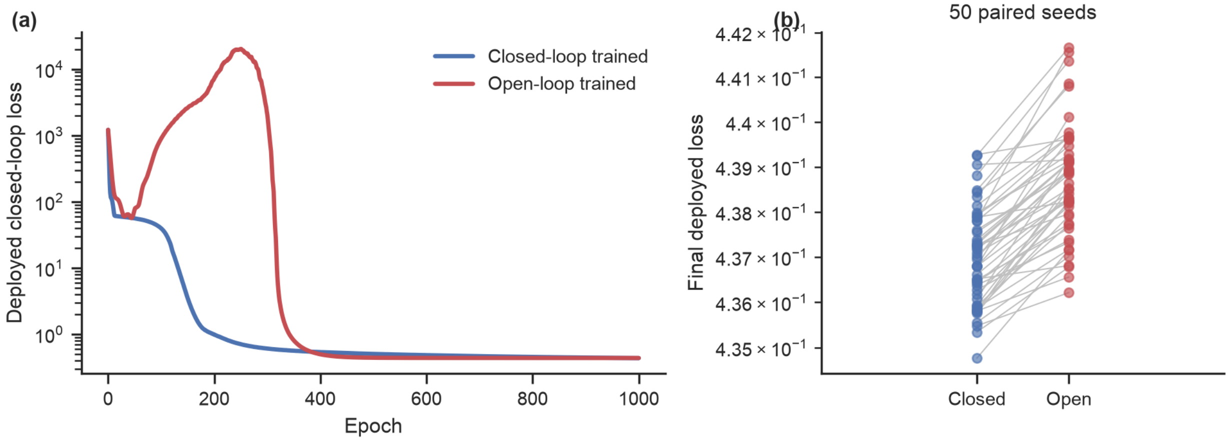

Figure 1: Core C1 reproduction under the original double-integrator setting. (a) Median deployed closed-loop test loss across paired seeds, with interquartile bands; the open-loop student shows the post-initial peak in 50/50 seeds. (b) Paired final deployed losses for the same initializations, showing that the open-loop students recover from the peak and finish with similar deployed losses.

Seeded executions of the original nonlinear notebook provide an independent artifact check: their observed ranges overlap the package runs for every behavioral metric, and both implementations show a transient peak followed by a final gap below 1 percent (Appendix H).

6 Result 2: Robustness and Tradeoff Audit

The robustness sweep contains 195 paired runs, with 15 seeds in each of 13 perturbation settings (Appendix C; Figure 2). The C1 peak appears in 165/195 runs: every finite SGD setting supports it, whereas none of the 15 Adam runs does. Final-gap support occurs in 60/195 runs, covering all 15 seeds at either altered learning rate, at the shorter horizon, and under Adam; in every other finite setting, the open-loop learner recovers by the end of training. Adam is thus the only condition with a persistent final gap and no transient peak.

All 15 `no_clip` runs become non-finite after the closed-loop learner diverges and supplies a non-finite imitation target to the open-loop student. Follow-up runs isolate the failure to closed-loop SGD: clipping only that learner rescues all 15 runs, and fully unclipped Adam remains finite, whereas lowering the SGD learning rate to 0.003 rescues none (Appendix F).

Spectral stages and coupled-stability crossings appear in all 180 finite robustness runs; the other 15 cannot be evaluated because unclipped training diverges. Loss-only stage detectors are more diagnostic-dependent: the derivative-style loss detector finds three-stage structure in many finite runs, but segmented changepoints on the original loss curves do not recover the same boundaries (Figure 7 in Appendix E).

The targeted A1 sweep covers six control penalties, with 20 seeds at each penalty and evaluation horizons $T_s = 10$ and $T_\ell = 200$ (Figure 3; Appendix B). All 120 runs meet both the union and radius criteria, and across runs a mean 47.24 percent of intervals with short-horizon improvement also show a radius increase. Sixty-two runs additionally meet the long-horizon loss-worsening criterion, always together with the radius signal. The other 58 do not meet the loss-worsening criterion, demonstrating that a radius increase is not sufficient for measurable loss worsening at the evaluated horizon.

At the interval level, the exclusive radius-only criterion is supported in 120/120 runs, compared with 0/120 for the exclusive loss-only criterion. These counts are separate from the run-level split: a run can contain radius-only intervals and still meet the loss-worsening criterion elsewhere in training.

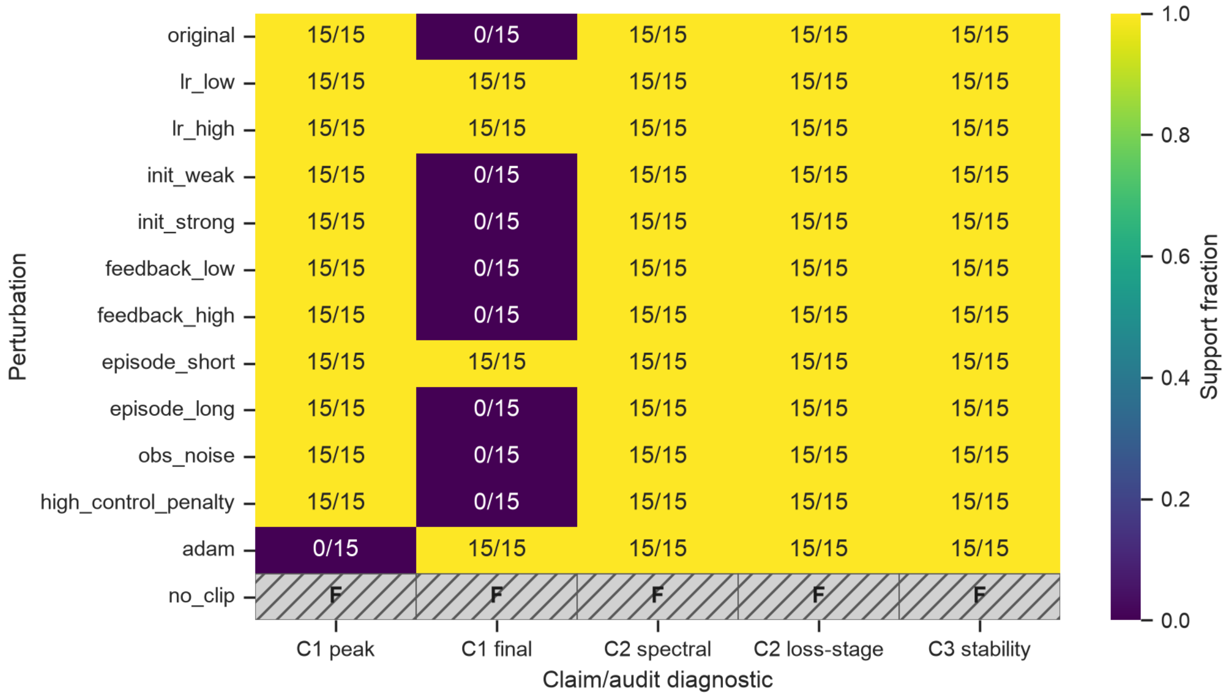

Figure 2: Robustness heatmap over perturbation settings and claim diagnostics. C1 has separate columns for the transient peak and final deployed-loss criteria; C2 has separate spectral and downstream loss-stage columns. The targeted multi-horizon A1 test appears in Figure 3. Non-finite, unclipped runs are marked F.

The balance between these run-level outcomes shifts with the control penalty. Counts for runs meeting the radius criterion but not the loss-worsening criterion are 5, 5, and 6 at $\beta = 0$, 0.001, and 0.005; they rise to 13, 13, and 16 at $\beta = 0.02$, 0.05, and 0.1. Yet the median final coupled radii are nearly identical in the two groups. Higher penalties are therefore associated with more runs that show the radius signal without meeting the behavioral-loss criterion. Across the threshold and horizon sensitivity grid, the invariant results are union and radius support in 120/120 runs and no loss-worsening support without a radius increase (Appendix E).

7 Result 3: Stage and Stability Diagnostics

The spectral detector reproduces C2 in all 50 core seeds, following the sequence defined in Section 4.3. Segmented regression on deployed loss gives a different account: its median boundaries fall at epochs 120 and 193, with segment slopes -0.0117, -0.0432, and -0.00065. Because none of the detector variants yields strict loss-based support for all three stages (Appendix E), C2 evidence comes from the spectral detector; the loss observer identifies different, downstream boundaries.

The original-setting stability transition also appears in all 50 seeds (Figure 4): every final coupled spectral radius is finite and below one, and the median crossing occurs at epoch 105.5. A derivative-style plateau detector places the plateau exit a median of one epoch from this crossing, whereas segmented loss changepoints fall at different downstream boundaries. The coupled-system spectrum therefore supplies the C3 diagnostic.

For linear tasks with linear, tanh, low-rank, or stacked controllers, the coupled transition matrix is available in closed form. Gated controllers and nonlinear tasks use local Jacobian spectra, obtained by differentiating the one-step map $(s_t, h_t) \mapsto (s_{t+1}, h_{t+1})$ along deployed rollouts. Where both constructions apply, their radii agree to within 10^{-5} at the origin of the linear and tanh double integrators. Across 10 core seeds evaluated with both methods, the along-trajectory Jacobian radius differs from the closed-form radius at the origin by

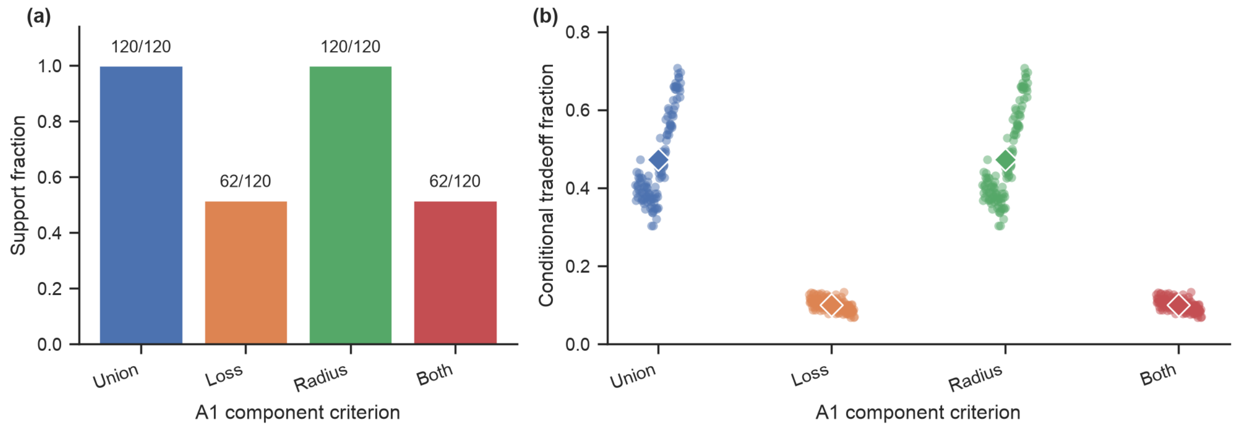

Figure 3: Targeted A1 component analysis over 120 paired tradeoff runs. (a) Support counts for the union criterion, long-horizon loss-worsening criterion, radius criterion, and the intersection of loss and radius criteria. (b) Seed-level conditional fractions among short-horizon improvement intervals, with diamonds marking means. The union and radius criteria are supported in 120/120 runs; loss worsening appears in 62/120 and never without the radius signal.

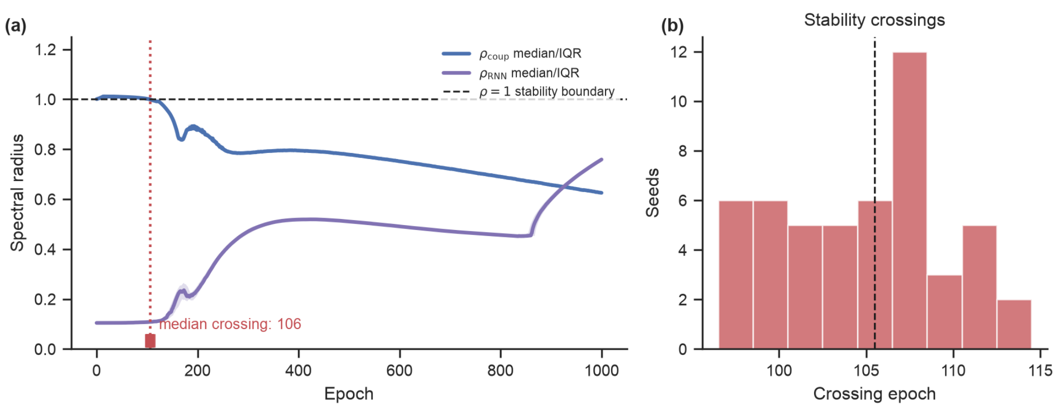

Figure 4: Coupled-system spectral analysis for the original setting. (a) Median coupled and RNN-only spectral radii across 50 seeds, with interquartile bands; the dashed line marks the $\rho = 1$ stability boundary and tick marks show per-seed crossing epochs. The plotted median marker is rounded to epoch 106, while the unrounded median used in the text is 105.5. (b) Distribution of stability-crossing epochs: the coupled radius crosses below one early in training, but the RNN-only radius shows no crossing aligned with that transition.

a median of only 3.6×10^{-7}; the median per-seed Pearson correlation rounds to 1.00 (Appendix I). Tracking uses its own coupled matrix and supports C3 in 3/10 runs, while Section 8.2 gives the GRU and ring results.

8 Result 4: Generalization Across Models and Tasks

8.1 Behavioral transfer

Among the 90 architecture and task extensions, 38 runs meet the final-loss criterion and 60 show the open-loop peak (Figure 6). The final-loss criterion is met in all 10 runs in each of the GRU, low-rank, and tracking settings, although the low-rank result arises from extremely large open-loop rollout losses. By contrast, all

10 tanh RNN runs show only the transient peak and finish with similar losses. Final-gap support is weaker on the moving-target rings: 5/20 GRU seeds and 2/20 tanh seeds meet the criterion, while peak support is 20/20 and 0/20, respectively.

The nearly memoryless fixed-target ring meets the final-loss criterion in only 1/10 seeds; mean final gaps also remain small in the harder moving-target variants, at 0.044 for GRU and 0.051 for tanh. Taken together, these results give strong support for final-loss transfer in the controller and tracking settings but weak support in the path-integration tasks.

8.2 Diagnostic (C3) transfer

Because behavioral and stability transfer need not coincide, we analyze them separately. Among variants with a closed-form diagnostic, C3 transitions appear in every tanh and low-rank run (10/10 each), whereas the task-specific coupled matrix supports the transition in 3/10 tracking runs. The GRU and ring variants instead use the local Jacobian described above, so we record their evidence separately from the closed-form C3 results: all 60 Jacobian runs register the prespecified persistent crossing, comprising 10 GRU double-integrator runs, 10 fixed-target ring runs, and 20 runs for each partial-observation controller.

Post-crossing radii and timing vary substantially: the GRU double integrator and fixed-target ring cross at median epochs 10 and 5 and finish at median radii 0.58 and 0.83, whereas the partial-observation variants cross later, at epochs 40 and 270, and finish near one with median radii 0.986 and 0.978. Individual seeds reach radii of 1.19; these variants also show weak behavioral transfer (Appendix I).

8.3 Scale and depth

Except for the size-64 GRU, the preceding experiments use single-layer controllers with hidden size 100. Using 10 paired seeds per condition, we first vary width with tanh controllers at sizes 50, 200, and 400 and a width-256 GRU. The depth test uses a two-layer tanh controller with 100 units per layer. A closed-form coupled matrix covers the tanh variants, including the stacked controller, whose linearization agrees with its automatic-differentiation Jacobian; the GRU-256 extension is evaluated behaviorally.

Width preserves the core signatures (Figure 5a). Every seed at every tanh width shows them (10/10 for each new size and 50/50 in the hidden-100 core), and every final gap remains below the threshold. The median peak/initial ratio rises steadily from 10.8 at hidden size 50 to 17.7 at size 100, 25.9 at size 200, and 50.4 at size 400, while median crossing epochs remain between 104 and 105.5. The hidden-256 GRU likewise retains final-loss divergence in all 10 seeds.

The two-layer controller trains stably and, in all 10 seeds, retains the spectral stages and crossing (median crossing epoch 92.5); its open-loop student also develops a persistent final gap in all 10 seeds (median 35 percent). Deployed loss rises again during training but stays below its high initial value, so no seed meets the strict peak criterion (Figure 5b). We did not repeat the full A1 sweep at each width and depth, leaving open whether the short/long-horizon tradeoff is unchanged across scales.

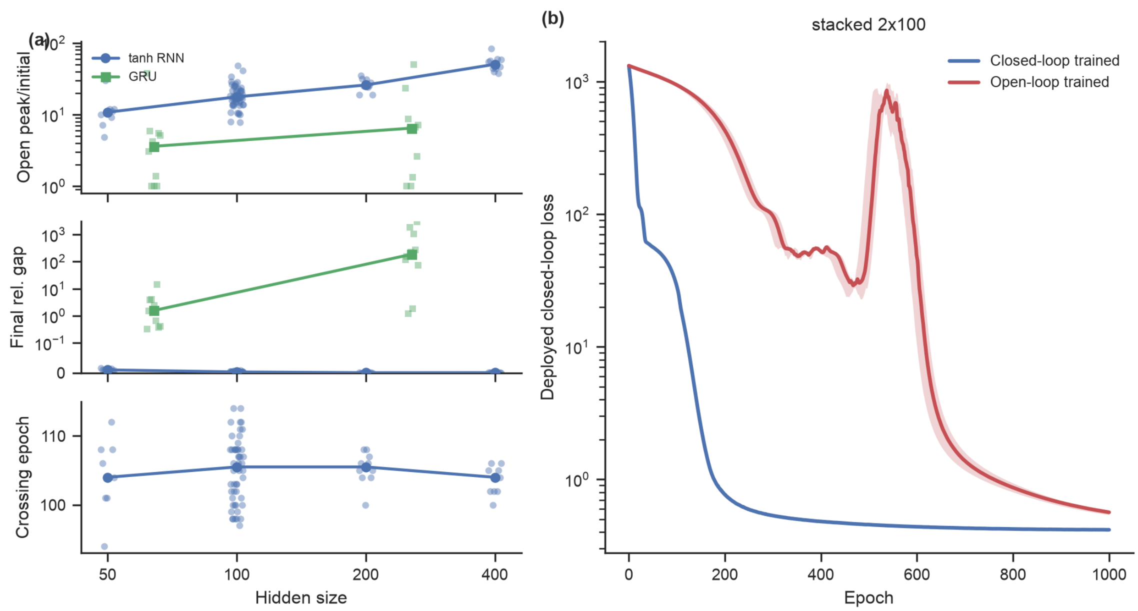

Figure 5: Scale and depth. (a) Peak/initial ratio, final relative gap, and stability-crossing epoch versus hidden size (10 paired seeds per size; hidden-100 is the 50-seed core; GRU at 64 and 256). (b) Median deployed loss for the two-layer stacked variant with interquartile bands. Loss rises again during training but remains below its initial value, then settles into a persistent final gap.

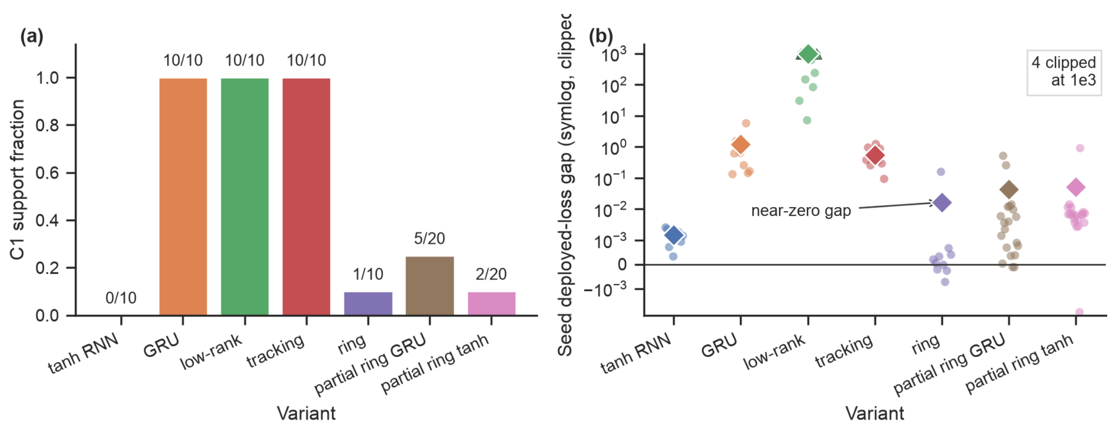

Figure 6: Generalization across architecture and task variants. Standard variants use $n = 10$ paired seeds; the partial-observation ring variants use $n = 20$. (a) C1 final-loss support fractions with exact counts. Tanh RNN final-loss support is 0/10, with peak support in 10/10 seeds. (b) Seed-level deployed-loss gaps with mean diamonds on a symmetric-log scale, clipped at 10^3 for display; triangles mark the clipped, extremely large low-rank rollout losses. The fixed-target ring provides a nearly memoryless sanity check.

9 Claim-Level Audit Matrix

Table 1 summarizes the evidence for original claims C1–C3 and audit questions A1–A2.

Table 1: Claim-level audit matrix for completed runs. C1–C3 are original claims under reproduction. A1–A2 are audit questions introduced here to test robustness, short/long horizon tradeoffs, and broader generalization.

Item	Evidence type	Original protocol	A1 audit	A2 audit	Lesson
C1: open/closed divergence	Final deployed loss and peak signature	Peak 50/50; final gap 0/50	Peak robust in finite SGD settings; final-gap diagnostic varies; no clipping fails	GRU/low-rank/tracking final gaps; tanh peak only; ring weak	Report deployed loss and peak signatures.
C2: stage structure	Spectral detector and loss-only observers	Spectral 50/50; segmented loss stages 0/50	Spectral C2 robust in finite linearizable settings	Transfers where a (local) linearization exists	Stage claims need diagnostics in the claim's own space.
C3: coupled stability	Coupled spectral radius and timing	Crossing 50/50; median epoch 105.5	Robust finite crossings	Closed form: tanh/low-rank/stacked; Jacobian crossings 60/60 for GRU/ring, with partial-observation rings ending near radius 1	Report the coupled spectrum and its timing vs stage boundaries.
A1: protocol robustness and tradeoff	Perturbation grid and short/long-horizon interval test	N/A	Union/radius 120/120; loss worsening 62/120; loss worsening without radius increase 0/120	N/A	Robustness claims need stress tests, explicit horizons, and component criteria.
A2: broader generalization	Architecture, task, scale, depth extensions	N/A	N/A	Behavioral: diagnostic-dependent; tracking strong, ring weak. Diagnostic: Sec. 8.2. Scale: signatures persist at hidden 50–400; depth: persistent gap, no strict peak	Effects depend on task structure, coupling, observability; report behavioral and diagnostic transfer separately.

10 Reporting Recommendations for Closed-Loop RNN Studies

The audit suggests four practical reporting choices.

- **Show the deployed trajectory and pair the comparison.** Report peak behavior as well as final performance, since transient deployment failures may disappear by the last epoch. Use the same initialization for both learners, then hold the training budget and evaluation rollout fixed across the two models.
- **Use a diagnostic that matches the claim.** Spectral stage claims need explicit eigenvalue criteria and boundary epochs, while loss changepoints should be identified as downstream observations. Stability claims should be tested on the coupled agent-environment system; when no closed-form transition matrix exists, local Jacobians provide a practical extension.
- **Document how the learner interacts with its environment.** Report the feedback strength and rollout horizon together with the location of gradient clipping and the teacher-input convention used for open-loop training.
- **Report failed runs and include a stress test.** Because average curves can hide unstable seeds or long plateaus, report failure counts beside aggregate performance.

11 Discussion

Together, the core experiments give a temporal account of C1–C3: the open- and closed-loop trajectories separate as the coupled system moves through the reported spectral stages, after which the open-loop learner recovers and finishes with a sub-threshold gap. Because the loss changepoints mark different boundaries, the stage evidence rests on the spectrum. The A1 study reveals a related asymmetry, with all 120 runs meeting the radius criterion but only 62 also meeting the long-horizon loss-worsening criterion. Behavioral transfer is strongest among the double-integrator architecture variants and on tracking; it is much weaker on the path-integration tasks.

For mechanistic RNN models in computational neuroscience, behavioral loss and dynamical stability provide different forms of evidence, and analyzing the full feedback loop reveals a transition absent from the RNN-only spectrum. Keeping these measures separate makes the proposed mechanism and its limits easier to see.

Scope: richer agent-environment interactions. Our evidence for this mechanism comes from low-dimensional environments that are locally linearizable around deployed trajectories, where training differentiates through rollouts and applies a dense cost at each step. In multi-step decision problems, the relevant coupled object may extend beyond the one-step map analyzed here to a long-horizon composition of transitions. Stochastic or non-differentiable reinforcement-learning loops fall outside the study, along with settings driven only by sparse rewards. The depth and partial-observation experiments probe parts of this boundary, while local Jacobians extend the diagnostic beyond systems with a closed-form transition matrix. A distinct use of spectral graph filtering over populations appears in evolutionary optimization (Ouyang et al., 2026).

Limitations. The transfer study covers a finite set of architectures; its path-integration evidence is limited to fixed- and moving-target ring tasks. A1 compares changes between evaluation intervals and therefore does not characterize the entire trajectory. Although local Jacobians make the gated controllers accessible, the partial-observation variants finish close to the stability boundary and sometimes above it. These experiments do not establish universal transfer or determine behavior in stochastic environments.

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A Artifact Entry Points

The public code repository contains the implementation, configurations, and saved outputs used for the tables and figures. Table 2 lists the main entry points and approximate runtimes.

Table 2: Artifact entry points. Full runtimes are approximate for one NVIDIA L40 GPU.

Component	File/script	Produces	Runtime	Expected output
Smoke validation	`scripts/run_smoke.sh`	Tiny raw outputs, tables, figures	< 10 min	`results/raw/smoke_*`
Full audit	`scripts/run_full_audit.sh`	Core, robustness, generalization, figures	16–18 h	sweep summaries, figures
Targeted C2/A1/A2	`scripts/run_targeted_c2_a1_a2.sh`	Tradeoff, hard ring, changepoints	8–10 h	targeted summaries
Extended analyses	`scripts/run_revision_experiments_fast.sh`	Scale/depth, Jacobian runs, gradient-clipping selectivity, convention ablation, artifact side-by-side	2 h (2 GPUs); 21–23 h sequential	extended-analysis summaries
Claim tables	`make_claim_tables` module	Claim matrix	< 1 min	`claim_matrix.csv`
Supplemental tables	`supplemental_audit_tables` module	Run accounting, C2 sensitivity, A1 split, peak/alignment checks	< 1 min	processed CSVs
Figures	`make_all_figures` module	Paper figures	< 1 min	`figure_*.png`

B Metric Definitions

Let s_t be the environment state, $o_t = Cs_t + \epsilon_t$ the observation, h_t the recurrent state, and $u_t = f_\theta(o_t, h_t)$ the action. Closed-loop deployment follows $s_{t+1} = F(s_t, u_t)$ and $h_{t+1} = G_\theta(o_t, h_t)$. The deployed loss is

$$L_{\text{deploy}}(\theta) = \frac{1}{T}\sum_{t=0}^{T-1}\left[\ell_s(s_t, s_t^\star) + \lambda_u\|u_t\|_2^2\right]. \tag{1}$$

The teacher-forced imitation loss compares learner and teacher actions on teacher-generated observations:

$$L_{\text{TF}}(\theta;\theta^\star) = \frac{1}{T}\sum_{t=0}^{T-1}\|f_\theta(o_t^\star, h_t) - f_{\theta^\star}(o_t^\star, h_t^\star)\|_2^2. \tag{2}$$

For linearizable double-integrator settings, we estimate an effective feedback gain at each recorded training epoch e from the state and action samples X_e and U_e: $\widehat{K}_e = (X_e^\top X_e + \alpha I)^{-1}X_e^\top U_e$. If $\mathcal{E}$ is the set of recorded epochs, the closed/open gain distance is $D_K = (\sum_{e\in\mathcal{E}}\|\widehat{K}_e^{\text{CL}} - \widehat{K}_e^{\text{OL}}\|_F^2)^{1/2}$.

For linear RNN controllers and linear environments, with $s_{t+1} = As_t + Bu_t$, $u_t = W_{ho}h_t$, and $h_{t+1} = (1-\eta)h_t + \eta(W_{hh}h_t + W_{ih}Cs_{t+1})$, the coupled transition matrix over $z_t = [s_t^\top, h_t^\top]^\top$ is

$$P_\theta = \begin{bmatrix} A & BW_{ho} \\ \eta W_{ih}CA & (1-\eta)I + \eta W_{hh} + \eta W_{ih}CBW_{ho}\end{bmatrix}. \tag{3}$$

We report $\rho_{\text{RNN}} = \max_i|\lambda_i(W_{hh})|$ and $\rho_{\text{coup}} = \max_i|\lambda_i(P_\theta)|$. For the closed-form series, the stability crossing is the first epoch that begins five consecutive epochs with $\rho_{\text{coup}} < 1$; local-Jacobian crossings use three consecutive finite probes, with Jacobians evaluated every five training epochs.

The open-loop peak signature is evaluated over the deployed-loss trajectory across training epochs. We count a post-initial peak when $\arg\max_e L_{\text{deploy},e}^{\text{OL}} > 0$ and $\max_e L_{\text{deploy},e}^{\text{OL}} > 1.5L_{\text{deploy},0}^{\text{OL}}$; we also report peak/initial and peak/final ratios.

The spectral detector follows the stage sequence defined in Section 4.3. A negative effective position gain together with a persistent unstable complex mode marks Stage 1; persistent entry of the coupled radius into the unit disk marks the start of Stage 2. Stage 3 is identified by growth of the largest-magnitude real eigenvalue after that crossing, which is the third mode in the original setting.

For comparison, the loss-only labels use smoothed log deployed loss $y_t = \log(\max(L_{\text{deploy},t}, \varepsilon))$. The segmented variant fits three piecewise-linear segments: the first must have a slope below -0.02, the second a slope magnitude below $\max(0.35|\text{slope}_1|, 0.004)$, and the third a slope below $\text{slope}_2 - 0.002$.

For A1, let I_s denote a short-horizon-improvement interval, $\Delta \log L_e^{(T_s)} < -\delta_{\text{short}}$. Long-horizon worsening is $I_\ell = [\Delta \log L_e^{(T_\ell)} > \delta_{\text{long}}]$, and a coupled-radius increase is $I_\rho = [\Delta\rho_{\text{coup},e} > 0]$. The reported components are

$$
\begin{aligned}
I_{\text{union}} &= I_s \wedge (I_\ell \vee I_\rho), & I_{\text{loss}} &= I_s \wedge I_\ell, \\
I_{\text{radius}} &= I_s \wedge I_\rho, & I_{\text{both}} &= I_s \wedge I_\ell \wedge I_\rho, \\
I_{\text{loss-only}} &= I_s \wedge I_\ell \wedge \neg I_\rho, & I_{\text{radius-only}} &= I_s \wedge I_\rho \wedge \neg I_\ell.
\end{aligned}
$$

A run meets a component criterion when at least three intervals satisfy it and those intervals account for at least 10 percent of the run's short-horizon-improvement intervals. In the targeted sweep, $T_s = 10$, $T_\ell = 200$, and $\delta_{\text{short}} = \delta_{\text{long}} = 0.005$.

For scalar paired metric M, the open/closed effect is $\Delta_i(M) = M_i^{\text{OL}} - M_i^{\text{CL}}$. We report the mean with a nonparametric bootstrap 95 percent CI and include seed-level SD or IQR where useful. Direction agreement is $N^{-1} \sum_i \mathbf{1}[\Delta_i(M) > 0]$, and the binary claim-support rate is $S(c) = N^{-1} \sum_i q_i(c)$.

For controllers without a single linear recurrent transition matrix, such as GRUs, and for tasks without a linear state-space form, such as the ring variants, we replace the closed-form coupled matrix with a local Jacobian. At probe points (s_t, h_t) collected along deployed rollouts, the one-step coupled map $f : (s_t, h_t) \mapsto (s_{t+1}, h_{t+1})$ is differentiated with parameters frozen and observation noise disabled. We summarize the resulting probe-level spectral radii by their median and retain the maximum as an additional diagnostic. Where both constructions apply, their radii agree at the origin to within 10^{-5} (Appendix I); local-Jacobian evidence is stored under the separate flag `claim_C3_stability_transition_jacobian` to distinguish it from closed-form C3 evidence.

C Audit Protocol Specification

Table 3: Mapping from original evidence to reproduction tests, distinguishing figure-level replication from claim-level decisions.

Original claim	Original evidence	Our test	Result	Decision
Closed/open divergence	Loss/gain figures and open-loop sharp peak	50 paired seeds plus peak-signature check	post-initial peak 50/50; final gap 0/50	Reproduced trajectory signature
Stage structure	Spectral stages: negative-position policy, stability phase, refinement	Spectral stage detector plus loss-only changepoints	spectral stages 50/50; segmented loss stages 0/50	Reproduced spectrally
Coupled stability	Coupled eigenvalue argument and Stage 2 alignment	Coupled radius crossing and spectral stage timing	50/50 crossings; median crossing 105.5	Reproduced
Motor-control applicability	Tracking extension	Tracking plus ring variants	final-loss support: tracking 10/10; ring family 8/50	Partial boundary

Table 4: Robustness perturbation grid. Its 15-seed baseline is separate from the 50-seed core reproduction, giving every grid row the same sample size.

Perturbation	Changed value	Seeds
`original`	Baseline double-integrator protocol	50 core; 15 grid
`lr_low`	learning rate 0.003	15
`lr_high`	learning rate 0.03	15
`init_weak`	recurrent initialization scale 0.03	15
`init_strong`	recurrent initialization scale 0.3	15
`feedback_low`	environment feedback strength 0.5	15
`feedback_high`	environment feedback strength 1.5	15
`episode_short`	rollout horizon 25 steps	15
`episode_long`	rollout horizon 100 steps	15
`obs_noise`	observation noise standard deviation 0.05	15
`adam`	optimizer Adam with learning rate 0.001	15
`high_control_penalty`	control penalty 0.05	15
`no_clip`	gradient clipping disabled	15

Table 5: Closed-loop RNN reproducibility audit protocol. Each row gives an audit step, its minimum implementation, and the output used for claim-level conclusions.

Audit step	Minimum requirement	Recorded output
Paired training	Clone the same initialization into open-loop and closed-loop learners; keep architecture and evaluation budget fixed.	Per-seed train loss, deployed closed-loop test loss, peak-over-training deployed loss, final loss, and effective-gain trajectory for both learners.
Seed-level uncertainty	Run enough seeds to estimate direction agreement and confidence intervals; report seed count explicitly.	Mean, 95 percent bootstrap CI, fraction of seeds supporting each claim, and seed-level failure/recovery rates.
Stage detection	Predefine detectors in the same diagnostic space as the claim; for this audit, separate loss-only stages from spectral stages.	Stage labels, plateau duration, transition epochs, spectral-crossing gaps, and frequency of detected stages across seeds/settings.
Coupled diagnostics	Compute RNN-only and coupled agent-environment spectra whenever the system can be linearized or locally linearized.	Spectral radius time series, stability-crossing epoch, plateau-exit gap, and association with recovery/failure.
Robustness perturbations	Change factors that alter optimization or environment coupling, such as optimizer, initialization scale, feedback strength, noise, horizon, control penalty, and clipping.	Claim support under each perturbation, effect-size changes, and perturbation-level failure rates.
Generalization check	Test at least one architecture or task variant that preserves action-dependent inputs and one that changes the recurrent/control structure.	Claim support by variant and a boundary-condition statement explaining when the phenomenon does or does not transfer.
Claim/audit decision	Classify each claim or audit question as reproduced, partially reproduced, or not reproduced using the same criteria across settings.	Matrix linking evidence type, original setting, robustness, generalization, and actionable lesson.

D Protocol Provenance

Table 6 divides settings into three provenance tiers. An *exact* setting has the same value and implementation as the public artifact; where the nonlinear notebook differs from the paper, *main-text-aligned* identifies the setting that follows the paper. Choices introduced here before the full sweeps are labeled *audit choice*, and the machine-generated setting-level map is `results/processed/provenance_map.csv`.

The 22 exact settings cover the double-integrator dynamics and controller as well as the training schedule. The task time step and feedback strength match the artifact; observation noise and the initial action are both zero. We retain its tanh RNN, using 100 hidden units with initialization scale 0.1 and leak 1. Training uses SGD at learning rate 0.01 for 1001 epochs, with batches of 100 and 50-step rollouts; gradients are clipped at 1.0. Loss is evaluated before the environment step and the control cost is summed, while the open-loop inputs are white noise.

Table 6: Main-text-aligned deviations and audit-specific choices. Setting-level notes, including the 22 exact matches, appear in `provenance_map.csv`.

Setting	Public notebook	This audit
Main-text-aligned (5)		
initial state, lower bound	-1	-2 (main text)
initial state, upper bound	1	2 (main text)
control penalty β	0.005	0 (main text)
state-cost reduction	mean over elements	squared Euclidean state cost, batch mean
loss normalization	summed over rollout	averaged over rollout ($1/T$)
Audit choices (10); original does not specify		
final-gap threshold	—	relative deployed-loss gap > 5 percent
peak criterion	—	peak $> 1.5\times$ initial, epoch > 0
spike criterion	—	peak $> 2.0 \times \max(\text{initial}, \text{final})$
spectral stage detector	—	gain < -0.01, persistence 5, largest-real-eigenvalue growth > 0.01
changepoint criteria	—	slopes -0.02 / $0.35\times$ / -0.002; min segment 20, stride 2
A1 horizons and δ	—	$[10, 50, 200]$; $\delta = 0.005$
seed counts	—	50 core / 15 robustness / 20 tradeoff / 10 generalization
evaluation protocol	—	fixed 10-episode eval batch, eval seed 12345
perturbation grid	—	Table 4 values
ridge gain α	—	10^{-8} in effective-gain regression

E Detector Threshold Sensitivity

Table 7 varies the final-gap threshold from 1 to 20 percent and preserves the setting-level counts. Every classification at the prespecified 5 percent threshold remains unchanged as the threshold increases, while lower thresholds add a few supporting seeds in three robustness conditions. The partial-observation tanh ring is the only clear boundary case: support falls from 19/20 seeds at 1 percent to 13/20 at 2 percent and 2/20 at 5 percent.

Table 7: C1 final-gap support counts as the relative deployed-loss threshold varies. The value prespecified before the full reported sweeps is 5 percent.

Setting	n	1%	2%	5%	10%	20%
original (core)	50	0	0	0	0	0
original (robustness grid)	15	0	0	0	0	0
adam / episode_short / lr_high / lr_low	15 each	15	15	15	15	15
feedback_low	15	3	0	0	0	0
obs_noise	15	3	2	0	0	0
init_strong	15	1	0	0	0	0
init_weak / feedback_high / episode_long / high_control_penalty	15 each	0	0	0	0	0
no_clip (all runs non-finite)	15	0	0	0	0	0
gru / low_rank	10 each	10	10	10	10	10
tracking_task	10	10	10	10	10	9
tanh_rnn	10	0	0	0	0	0
ring_path_integration	10	1	1	1	1	1
ring_partial_obs_gru	20	10	8	5	2	2
ring_partial_obs_tanh	20	19	13	2	1	1

For A1, we recompute the component criteria over $\delta \in \{0.0025, 0.005, 0.01, 0.02\}$ and horizon pairs $\{(10, 200), (10, 50), (50, 200)\}$. Every grid cell has union and radius support in 120/120 runs, with no loss-only support (Table 8). A smaller δ labels more intervals as short-horizon improvements and changes the fraction that also show long-horizon worsening; a larger δ leaves fewer qualifying intervals. The grid-invariant union and radius results determine the A1 conclusion, while the loss-worsening count is reported as threshold-dependent.

Table 8: A1 sensitivity grid. Cell entries are runs (of 120) that meet the long-horizon loss-worsening criterion: at least three qualifying intervals comprising at least 10 percent of short-horizon-improvement intervals. In every cell the union and radius criteria are 120/120 and the exclusive loss-only criterion is 0/120. The setting prespecified before the full reported sweeps is $\delta = 0.005$ with horizons $(10, 200)$.

δ	$(T_s, T_\ell) = (10, 200)$	$(50, 200)$	$(10, 50)$
0.0025	2	0	0
0.005	**62**	29	0
0.01	107	99	0
0.02	117	25	0

Across the eight loss-based detector variants in Table 9, changes to the segmentation settings or smoothing leave the fitted boundaries nearly invariant: τ_1 lies between 110 and 120 and τ_2 between 192 and 194, consistent with the medians in Section 7. At the main threshold, the first-segment slope of -0.0117 misses the -0.02 fast-descent cutoff, although looser slope thresholds yield Stage-1 support in 47–50/50 seeds. No variant detects the Stage-2 slow phase or Stage-3 reacceleration, leaving strict three-stage support at 0/50 throughout.

Table 9: C2 sensitivity of the segmented loss-only changepoint observer on 50 original-setting seeds, including segmentation variants and component-level support. Spectral C2 support is 50/50; strict loss-based three-stage support is 0/50 for every variant.

Detector variant	**Stage-1 fast**	**Strict three-stage**	**Median τ_1**	**Median τ_2**
Main segmented	0/50	0/50	120	193
Loose segmented	47/50	0/50	120	193
Strict segmented	0/50	0/50	120	193
Short-segment binary style	0/50	0/50	120	193
Very loose reacceleration	50/50	0/50	120	193
Stride-1 segmentation	0/50	0/50	119	192.5
Min-segment-40 segmentation	0/50	0/50	110	194
Smoothed log-loss (window 9)	0/50	0/50	118	192

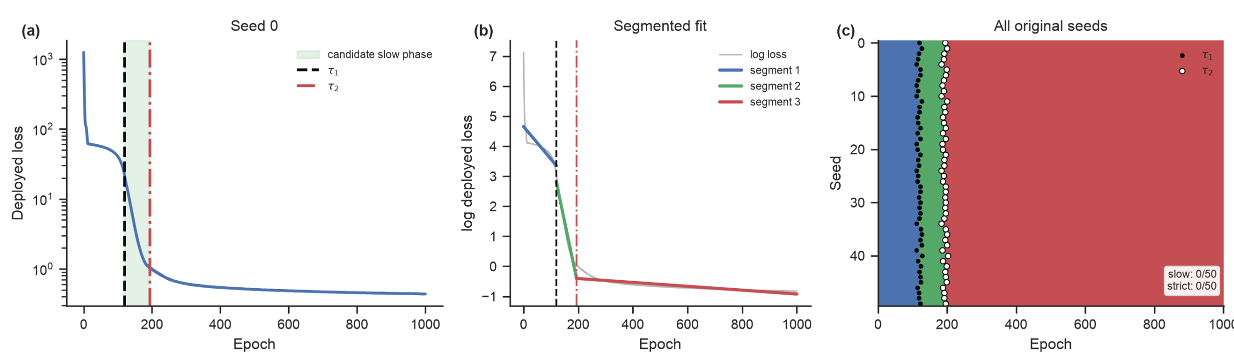

Figure 7: Stage analysis using downstream loss changepoints. Panels (a,b) show a representative seed with boundaries τ_1 and τ_2 and segmented fits; panel (c) shows the aggregate raster for 50 seeds, with black and white dots marking τ_1 and τ_2. The detector uses minimum segment length 20, Stage-1 slope < -0.02, Stage-2 slope magnitude below $\max(0.35|\text{slope}_1|, 0.004)$, and Stage-3 slope $< \text{slope}_2 - 0.002$. Strict three-stage support is 0/50.

F Gradient-Clipping Failure Analysis

Failure sequence. In all 15 unclipped runs, the closed-loop learner fails first: training and deployed losses become non-finite by epochs 5–7 and 4–6 as the coupled radius grows from 1.000–1.045 until it becomes non-finite. Its output then makes the open-loop student's training signal non-finite from the first epoch; per-seed details are in `no_clip_forensics.csv`.

Learner and optimizer dependence. Across 15 seeds per condition, 0/15 runs remain finite when only the closed-loop learner is unclipped; when only the open-loop learner is unclipped, all 15 remain finite and show the peak-and-recovery signature. Fully unclipped Adam remains finite in 15/15 cases, with peaks in 6, whereas fully unclipped SGD at learning rate 0.003 remains finite in 0/15. In these tested conditions, clipping is required for the closed-loop SGD learner but not for the open-loop learner or Adam.

G Open-Loop Teacher-Input Convention Ablation

In all 15 ablation seeds, teacher-rollout inputs produce a persistent final deployed-loss gap and no post-initial peak; the median relative-gap ratio, $(L^{\mathrm{OL}} - L^{\mathrm{CL}})/L^{\mathrm{CL}}$, is approximately 147. Under the white-noise convention, all 65 comparable runs show a peak before finishing below the 5 percent final-gap threshold; per-seed values are in the `robustness_open_loop_teacher_rollout` summary.

H Original-Artifact Side-by-Side Comparison

Table 10 compares ten package runs with five executions of the original `train_non_linear` notebook, for which we inject seeds because the notebook sets none. The observed ranges overlap for every behavioral quantity in the table, and the notebook shows a large transient open-loop peak with a final gap of 0.0005, below 1 percent. Coupled-radius metrics are available only for the package because the notebook does not compute them.

Table 10: Original-artifact notebook versus package replication under the notebook protocol. Values are means with seedwise [min, max].

Metric	**Notebook (5 seeds)**	**Package (10 seeds)**
Closed final deployed loss	0.061 $[0.047, 0.073]$	0.059 $[0.058, 0.060]$
Open final deployed loss	0.062 $[0.047, 0.075]$	0.060 $[0.059, 0.061]$
Open-loop peak epoch	213.2 $[212, 214]$	212.2 $[202, 218]$
Open peak/initial ratio	47.9 $[27.3, 80.0]$	39.8 $[15.5, 68.9]$
First coupled-stable epoch	—	103.1 $[98, 112]$

I Validation of Local-Jacobian Spectra and Transfer Across Tasks

Validation. On linear and tanh double-integrator systems, the Jacobian spectral radius at the origin matches the closed-form coupled radius to within 10^{-5}; the test suite applies the same comparison to the two-layer stacked closed form and its automatic-differentiation Jacobian. Across 10 core-protocol seeds evaluated with both methods, the median absolute difference along the trajectory is 3.6×10^{-7} and the maximum is 5×10^{-3}; the median per-seed Pearson correlation rounds to 1.00 (Figure 8e).

Transfer trajectories. Figure 8 shows median local-Jacobian coupled-radius trajectories with per-seed crossing ticks. Median crossings occur at epochs 10 and 5 for the GRU double integrator and fixed-target ring, whose final radii are 0.58 and 0.83. For the partial-observation variants, the corresponding crossings occur at epochs 40 and 270 and the final radii are 0.986 and 0.978; individual seeds reach 1.19, consistent with their weak behavioral transfer.

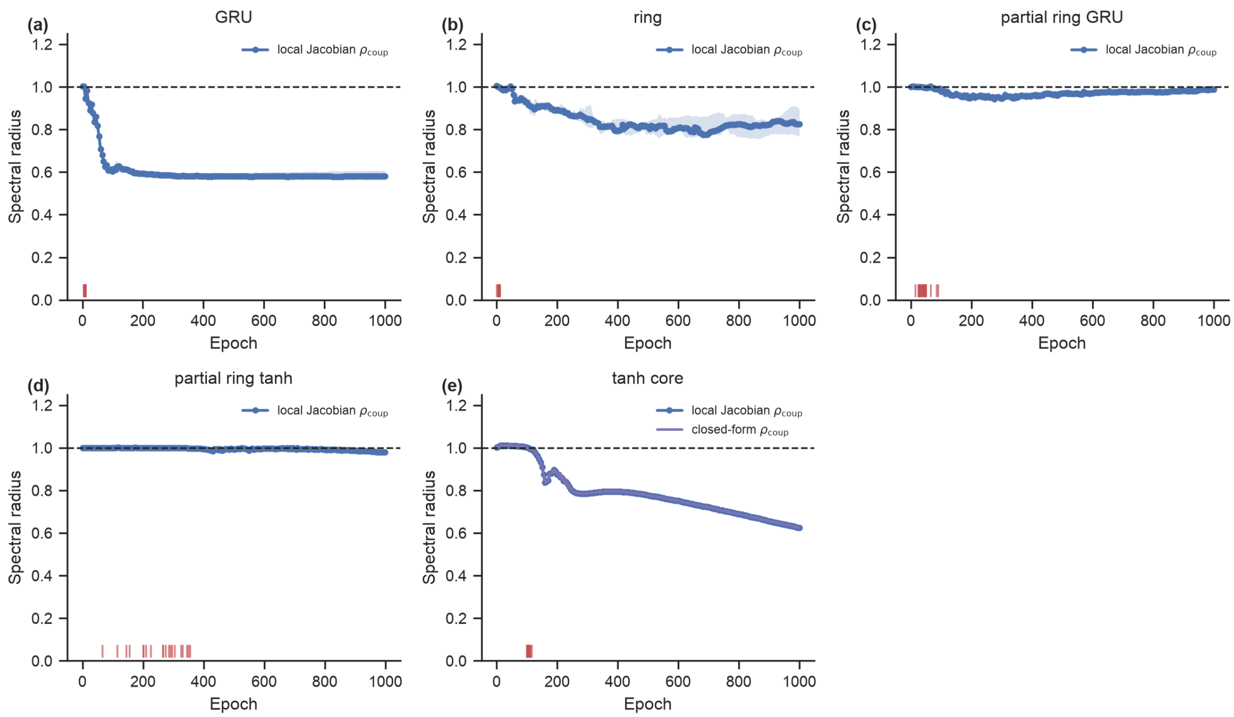

Figure 8: Local-Jacobian coupled spectral radius over training for the diagnostic-transfer runs. Curves show the median with an interquartile band over seeds; Jacobians are evaluated every five training epochs, the dashed line marks $\rho = 1$, and ticks mark per-seed crossing epochs. Panels show (a) the GRU double integrator, (b) ring path integration, (c,d) the partial-observation ring with GRU and tanh controllers, and (e) the tanh core protocol, where the along-trajectory Jacobian radius overlaps the closed-form radius at the origin.